

Standard Form & Indices

Test 3 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $2^5 \times 2^4$ as a single power of 2.

[1 mark]

A2. Write $6^8 \div 6^5$ as a single power of 6.

[1 mark]

A3. Write $(4^2)^3$ as a single power of 4.

[1 mark]

A4. Work out the value of $5^1 \times 5^2$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 7^0 .

[1 mark]

B2. Work out the value of 3^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 10^{-5} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 83,000 in standard form.

[2 marks]

C2. Write 9,100,000 in standard form.

[2 marks]

C3. Write 0.0025 in standard form.

[2 marks]

C4. Write 0.00006 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 5.1×10^3 as an ordinary number.

[1 mark]

D2. Write 6×10^{-4} as an ordinary number.

[1 mark]

D3. Work out $(2 \times 10^3) \times (4 \times 10^4)$. Give your answer in standard form.

[3 marks]

D4. Work out $(6 \times 10^6) \div (3 \times 10^2)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 2^7 .

[1 mark]

E2. $\sqrt{110}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $m^7 \times m^2$.

[1 mark]

F2. Simplify $n^9 \div n^5$.

[1 mark]

F3. Simplify $(k^4)^2$.

[1 mark]

F4. Simplify $3x^2 \times 5x^4$.

[2 marks]

F5. Simplify $(2y^4)^3$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $25^{1/2}$.

[1 mark]

G2. Work out the value of $64^{1/3}$.

[1 mark]

G3. Work out the value of $32^{2/5}$.

[2 marks]

G4. Work out the value of $4^{-3/2}$.

[2 marks]